

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2-DEBUGGING&OPTIMIZATION TASK

CourseCode:	CSA0305	CourseTitle:	Data Structures
CO Assessed:	CO2-Manipulation(BL3,BL4)	Assessment:	Assessment Tool 2- Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	M. VAISHNAV	Reg. No.:	192524251
Section/Batch:	2025	Date:	24/07/2026

Code of Conduct

I M. VAISHNAV (192524251)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. VAISHNAV (192524251)

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answers using Scenario-Based, in evaluation the answers should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section/Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1- 5	25
LinkedList Data Structure	List Operations	6-10	25
Queue Data Structures	Queue Operations	11-15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16-20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack - Balancing Symbols
5 marks**Q6**
Singly Linked List
5 marks**Q7**
Stack
5 marks**Q1 Binary Search**

5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int n, int key) {  
2     int low = 0, high = n - 1;
```

Test Input

stdin input (optional)

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack - Balancing Symbols
5 marks**Q6**
Singly Linked List
5 marks**Q7**
Stack
5 marks**Q8**
Linked List
5 marks**Q9**
Linked List

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop() {  
2     if(top == -1) {  
3         printf("Underflow");  
4         return -1;  
5     }  
6     return stack[top--];  
7 }
```

Test Input

stdin input (optional)

Output

[Run](#)[Save](#)[Submit Fix](#)

Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int stack[5];
2 int top;
3
4 void init() {
5     top = -1; // -1 indicates the stack is empty
6 }
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List

Identify the errors in the following snippet.

```
if(ch == ')') {
    while(stack[top] != '(') {
        postfix[j++] = pop0;
    }
    pop0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if(ch == ')') {
2     while(stack[top] != '(') {
3         postfix[j++] = pop();
4     }
5     pop(); // remove the '(' from the stack
6 }
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List

What is the issue?

```
temp = head
while temp.next != NULL:
    print(temp.data)
temp = temp.next
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp = head
2 while temp != None:
3     print(temp.data)
4     temp = temp.next
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List

Point out the error.

```
POP(stack):
if top == 0:
    print("Underflow")
else:
    return stack[top--]
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 def POP(stack):
2     if top == -1:
3         print("Underflow")
4         return None
5     else:
6         x = stack[top]
7         top -= 1
8         return x
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):  
if front == -1:  
    print("Underflow")  
else:  
    return Q[front++]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q):  
2 if front > rear:           // queue is empty  
3     print("Underflow")  
4     return  
5 else:  
6     x = Q[front]  
7     front = front + 1  
8     if front > rear:       // became empty now  
9         front = -1  
10        rear = -1  
11    return x
```

Test Input

stdin input (optional)

Output

```
Compilation Error:  
C:\Windows\TEMP\sandbox_192524251  
RA_6a7bfba3e8ebc5.58420434\main.c  
:1:1: warning: return type  
defaults to 'int' [-Wimplicit-  
int]
```

Run

Save

Submit Fix

Q5 ✓
Doubly Linked List
5 marks

Q6 ✓
Queue
5 marks

Q7 ✓
Queue - Deque
5 marks

Q8 ✓
Binary Search
5 marks

Q9 ✓
Circular Queue
5 marks

Q10 ✓
Stack - Postfix Expression
5 marks

10/10 answered

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <cctype>
2 #include <string>
3 #include <stack>
4
5 int evaluatePostfix(const std::string& expression) {
6     std::stack<int> st;
7
8     for (size_t i = 0; i < expression.size(); i++) {
9         char ch = expression[i];
10
11         if (ch == ' ' || ch == '\t') continue;
12
13         if (isdigit(ch)) {
14             st.push(ch - '0'); // digit operand
15         } else {
16             int op2 = st.top(); st.pop();
17             int op1 = st.top(); st.pop();
18
19             int result = 0;
20             switch (ch) {
21                 case '+': result = op1 + op2; break;
22                 case '-': result = op1 - op2; break;
23                 case '*': result = op1 * op2; break;
24                 case '/': result = op1 / op2; break;
25             }
26             st.push(result);
27         }
28     }
29     return st.top();
30 }
```

Test Input

stdin input (optional)

Output

Compilation Error:
C:\Windows\TEMP\sandbox_192524251
RA_6a72f77f372425.44975094\main.c
:1:10: fatal error: cctype: No
such file or directory
1 | #include <cctype>

🔴 Pending manual review